

SATHVIKA KOTTHA

Sr. DATA ENGINEER

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PROFESSIONAL SUMMARY

- Senior Big Data Engineer/Python ETL Developer with over 7 years of IT industry experience, including 5+ years in Big Data and Cloud solutions.
- Expertise in transforming business problems into Big Data solutions and developing comprehensive Big Data strategies and roadmaps.
- In-depth understanding of Hadoop architecture and components, including MapReduce, HDFS, HBase, Hive, Sqoop, Pig, Zookeeper, and Flume.
- Skilled in data processing with Spark, converting Hive/SQL queries into Spark transformations using Spark Data Frames and Scala.
- Experience with Azure Cloud, Azure Data Factory, Azure Data Lake Storage, Azure Synapse Analytics, Azure Analytical Service, Apache Spark, and Databricks.
- Proficient in real-time data processing using Kafka, Kafka brokers, and integrating Apache Kafka with Apache Storm for real-time data pipelines.
- Hands-on experience with AWS cloud services, including EC2, S3, Redshift, EMR, CloudWatch, and automating data workloads to Snowflake Warehouse.
- Experience in data migration from RDBMS to cloud data warehouses, including Azure Data Lake, Azure SQL Database, and Snowflake.
- Strong scripting skills using Python API, PySpark API, and Spark API for data analysis and ETL pipeline development.
- Designed and optimized Hive tables with partitions and bucketing, both Managed and External, for enhanced performance.
- Developed and maintained business applications across financial, retail, and healthcare domains, leveraging Hadoop and cloud technologies.
- Used Team Center for cross-functional collaboration by offering role-based access, secure data sharing, and project management tools.
- Good understanding of NoSQL databases like MongoDB, Azure, PostgreSQL, HBase, and Cassandra.
- Proficient in writing Azure PowerShell scripts to move data from local file systems to HDFS Blob storage.
- Experienced with ETL TALEND Data Fabric components, using features like context variables and integrating MySQL, Oracle, and Hive databases.
- Proven ability to analyze data using HQL, Pig Latin, and custom MapReduce programs, designing star schema data warehouses for efficient data organization and analysis.

TECHNICAL SKILLS:

Big Data Tools: Hadoop, Map Reduce, Pig, Hive, YARN, Kafka, Flume, Sqoop, Impala, Oozie, Airflow, Zookeeper, Spark, Ambari, Mahout, MongoDB, Cassandra, Avro, Storm, Parquet and Snappy.

Hadoop Distributions: Cloudera (CDH3, CDH4, and CDH5), Hortonworks, MapReduce and Apache.

Languages: Python, SQL, Scala, XML

Cloud Platforms: AWS, Azure.

No SQL Database: Cassandra, MongoDB and HBase

Python Libraries: NumPy, Matplotlib, NLTK, Statsmodels, Scikit-learn/sklearn, SOAP

XML Technologies: XML, XSD, DTD, JAXP (SAX, DOM), JAXB

Operating Systems: Sun Solaris, HP-UNIX, RedHat Linux, Ubuntu Linux, and Windows XP/Vista/7/8
ETL /Data Warehouse Tools: Informatica, Airflow, Talend, Snowflake, Redshift, HANA, Teradata, Exasol
DB Languages: MySQL, PL/SQL, PostgreSQL and Oracle
CI/CD: Jenkins, GitLab, Jira, Confluence

PROFESSIONAL EXPERIENCE:

Senior Data Engineer
Fidelity Investments - Durham, NC

July 2022 – Present

Project:

The project involved developing Hadoop applications in Azure, managing ETL pipelines using tools like Hive, Pig, and Kafka, and orchestrating workflows with Azure Data Factory and Airflow. Key tasks included optimizing dashboards, managing Hadoop clusters, and using Spark for data transformation and migration.

Responsibilities:

- Developed and deployed custom Hadoop applications in an Azure environment.
- Created and managed workflows in Oozie to schedule jobs on Hadoop clusters for daily, weekly, and monthly batch cycles.
- Configured Hadoop tools including Hive, Pig, Zookeeper, Flume, Impala, and Sqoop.
- Deployed initial Azure components such as Azure Virtual Networks, Azure Application Gateway, Azure Storage, and Affinity Groups.
- Created ETL mappings for operational dashboards to track KPIs and business metrics, enabling detailed drill-down reports.
- Worked with Big Data Analytics using Hadoop, Hive, MapReduce, OLAP databases, HP Vertica, and Tableau.
- Developed context variables and groups in Talend to run jobs across different environments.
- Created and managed Azure SQL databases, including monitoring, restoring, and migrating from Microsoft SQL Server.
- Designed and optimized Tableau reports according to customer requirements.
- Orchestrated data workflows using Azure Data Factory, Airflow 1.8, and Airflow 1.10 on multiple cloud platforms.
- Developed Kafka consumer APIs in Scala for consuming data from Kafka topics.
- Designed and implemented data pipelines using Data Fabric jobs with Sqoop, Spark, Scala, and Kafka, and managed data from Oracle and MySQL servers.
- Wrote Spark programs to move data from storage locations, including data loading, validation, and transformation.
- Designed efficient data models utilizing Hive complex datatypes and the Parquet file format for large-scale queries.
- Used Cloudera Manager for continuous monitoring and managing Hadoop clusters, including OS installation, Hadoop updates, and version upgrades.
- Contributed to deploying Siemens Team Center 10.0 across multiple divisions and plants.
- Managing documentation across the lifecycle, from requirements specifications and design documents to manuals and compliance reports using Siemens Team Center.
- Developed Spark applications using PySpark and Spark-SQL for data extraction, transformation, and aggregation from various file formats.
- Utilized Scala functions, dictionaries, and data structures for better code reusability and efficiency.
- Employed Spark SQL API in PySpark for data extraction, loading, and SQL queries.
- Involved in data migration processes using Azure, integrating with GitHub repositories and Jenkins.
- Used Spark Data Frames Operations for data validation and analytics on Hive data, and processed data in Azure services such as Azure Data Lake and Azure SQL.
- Experienced in orchestrating data pipelines using Apache Airflow to interact with services like Azure Data-bricks, Azure Data Factory, Azure Data Lake, and Azure Synapse Analytics.

- Monitored Hadoop jobs using Hadoop Resource Manager.
- Developed data ingestion pipelines on Azure HDInsight Spark cluster with Azure Data Factory and Spark SQL and worked with Cosmos DB (SQL API and Mongo API).
- Extensively used Azure Data Factory for data transformations, Integration Runtimes, Azure Key Vaults, triggers, and migrating pipelines using ARM Templates.
- Developed ETL processes using DataStage Open Studio to load data into HDFS via Flume and Sqoop and performed structural modifications with MapReduce and Hive.
- Created Spark scripts and UDFs using Spark DSL and Spark SQL for data aggregation, querying, and writing data back into RDBMS through Sqoop.
- Implemented MapReduce jobs using Java API, Pig, and Hive for data extraction, transformation, and aggregation from various file formats, including Parquet, Avro, XML, JSON, CSV, and ORCFILE.
- Consolidated data into a Data Lake from multiple source databases such as Teradata using Hadoop technologies like Sqoop and Hive/HQL.

Environment: Azure, Kafka, Map Reduce, Python, Data Proc, Data Storages, SAS/ETS, Hadoop, Hive, Pig, Spark, PySpark, spark, SQL, Azure SQL DW, Data brick, Siemens Team Center 10.0, Azure Synapse, Azure Data Lake, ARM, Azure HDInsight, Blob storage, Oracle 12c, Cassandra, Git, Zookeeper, Oozie, Data bricks.

Senior Big Data Engineer
Nationwide, Columbus, OH

Sep 2020– June 2022

Project:

The project involved processing and transforming healthcare and consumer data using AWS and Hadoop tools. Key tasks included migrating data to Snowflake, building ETL pipelines for Redshift, and integrating Kafka for real-time processing. Technologies like Spark, Hive, Pig, Python, and AWS Lambda were used for data transformation and reporting.

Responsibilities:

- Worked with Spark-Streaming APIs to transform and process data from Kafka.
- Used Apache NIFI to decompress and move JSON files from local storage to HDFS.
- Implemented installation and configuration of a multi-node cluster on AWS EC2.
- Handled AWS Management Tools like CloudWatch and CloudTrail.
- Migrated data from RDBMS to Snowflake cloud data warehouse with strong experience.
- Installed and configured Apache Airflow for AWS S3 bucket and created DAGs to run in Airflow.
- Performed ETL testing by running jobs, extracting data with queries, transforming it, and uploading to data warehouse servers.
- Built a Big Data analytical framework for processing healthcare data for medical research using Python, Java, Hadoop, Hive, and Pig. Integrated R scripts with MapReduce jobs.
- Created data pipelines for ingestion, aggregation, and loading consumer response data into AWS Redshift for Tableau feeds.
- Developed data warehouse model in Snowflake for over 100 datasets using WhereScape, creating reports in Lower based on Snowflake connections.
- Used Siemens Team Center for supporting the custom workflows and process automation to streamline approval cycles, change management, and compliance tracking.
- Developed stored procedures and views in Snowflake and used them in Talend for loading dimensions and facts.
- Stored log files in AWS S3, utilizing versioning for highly sensitive information.
- Integrated AWS DynamoDB with AWS Lambda to store item values and back up DynamoDB streams.
- Developed a Python application for Google Analytics data aggregation and reporting.
- Loaded Salesforce data every 15 minutes on an incremental basis to BigQuery raw and UDM layer using SOQL, Hive, Spark, Scala, Python, and Shell Script.
- Worked as a Program Manager with a Technical Architect to design, deploy, and implement solutions.

- Designed and implemented a configurable data delivery pipeline for scheduled updates to customer-facing data stores built with Python.
- Ingested data through cleansing and transformations, leveraging AWS Lambda, SAS, Glue, and Step Functions.
- Stored data in AWS S3 like HDFS and performed EMR programs on the stored data.
- Used AWS-CLI to suspend an AWS Lambda function and to automate backups of ephemeral data-stores to S3 buckets and EBS.
- Transferred data using Informatica tool from AWS S3 to AWS Redshift.
- Wrote MapReduce code using Python to address certain security issues in the data.
- Integrated data sources from Kafka (Producer and Consumer API) for data stream-processing in Spark using AWS Network.
- Designed data models for data-intensive AWS Lambda applications aimed at complex analysis.
- Created analytical reports for traceability, lineage, and key business elements from Aurora.
- Set up the entire app stack and configured Logstash to send Apache logs to AWS Elasticsearch.
- Developed Spark code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- Implemented various AWS computing and networking services to meet application needs.
- Wrote HiveQL as per the requirements and processed data in the Spark engine, storing it in Hive tables.
- Developed a Python script to transfer data from on-premises to AWS S3.

Environment: Hadoop, HDFS, MapReduce, Snowflake, Scala, Data bricks, SAS Data Integration Studio, Information Map Studio, Siemens Team Center ,Management Console. SAS OLAP & EBI Server, Teradata, SQL Server, Postgres SQL, Spark, Impala, Hive, MongoDB, Pig, HBase, Oozie BigQuery, Data Proc, Data Storage, Sqoop, Flume, Oracle, AWS, MySQL, SQL Server, Python, Spark.

Big Data Engineer/ Data Migration Nike - Oregon

Feb 2019 – Aug 2020

Project:

The project involved automating ETL processes, migrating databases to the cloud, and building real-time data pipelines using Spark-Streaming and Kafka. Data was ingested from AWS S3 and Redshift, transformed with Spark SQL and Hive, and visualized in Tableau. Performance optimization and cloud integration were key focus areas.

Responsibilities:

- Experience in Job management using Fair scheduler and developed job processing scripts using Oozie workflow.
- Prepared scripts to automate the ingestion process using PySpark and Scala as needed through various sources such as API, AWS S3, Teradata, and Redshift.
- Migrated databases to cloud platforms and performed performance tuning.
- Used Spark and Hive for implementing the transformations needed to join the daily ingested data to historic data.
- Used Spark-Streaming APIs to perform necessary transformations and actions on the fly for building the common learner data model which gets the data from Kafka in near real-time.
- Built different visualizations and reports in Tableau using Snowflake data.
- Experienced in performance tuning of Spark applications for setting the right batch interval time, correct level of parallelism, and memory tuning.
- Implemented MDM solutions and real-time enterprise application integration.
- Developed a reusable framework for future migrations that automates ETL from RDBMS systems to the Data Lake utilizing Spark Data Sources and Hive data objects.
- Experience in designing Star schema and Snowflake schema for Data Warehouse and ODS architecture.
- Designed BI Applications in Tableau, Power BI, SSIS, SSRS, SSAS, Informatica, and Data Stage.
- Worked on AWS Data pipeline to configure data loads from S3 to Redshift.
- Migrated on-premise database structure to Redshift data warehouse.

- Used AWS Redshift to extract, transform, and load data from various heterogeneous data sources.
- Built data pipelines using PySpark and Spark SQL.
- Produced Pytest to ensure functionality while building Python wrappers to get all feature functionality created in Spark for analytical products.
- Developed ETL jobs in AWS Glue using Scala and Python to migrate existing Informatica mappings and workflows to Glue jobs and ingest data from several feeds to the data lake.
- Optimized existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frames, and Pair RDD's.
- Built pipelines to move hashed and un-hashed data from XML files to Data Lake.
- Expertise in analyzing data using Pig scripting, Hive Queries, Sparks (Python), and Impala.

Environment: Spark, Kafka, Hadoop, HDFS, AWS, Spark-SQL, Snowflake, Python, Map Reduce, Pig, Hive, Oracle 11g, My SQL, MongoDB, HBase, Oozie, Zookeeper, MDM, Tableau.

Hadoop Developer **L&T – Pune, India**

June 2015 – June 2017

Description: Larsen & Toubro Limited, abbreviated as L&T, is an Indian multinational conglomerate, with interests in industrial technology, heavy industry, engineering, construction, manufacturing, power, information technology, military and financial services. It is headquartered in Mumbai, Maharashtra.

Responsibilities:

- Involved in various phases of development, analyzed and developed the system using Agile Scrum methodology.
- Responsible for data extraction and data ingestion from different data sources into Hadoop Data Lake by creating ETL pipelines using Pig and Hive.
- Built pipelines to move hashed and un-hashed data from XML files to Data Lake.
- Extensively worked with Spark-SQL context to create data frames and datasets to pre-process the model data.
- Experienced in writing live real-time processing core jobs using Spark Streaming with Kafka as a data pipeline system.
- Created groups and users in Tableau Server.
- Worked on migrating MapReduce programs into Spark transformations using Spark and Scala.
- Worked with EMR and set up the Hadoop environment in AWS EC2 Instances.
- Developed Spark jobs using Scala on top of Yarn/MRv2 for interactive and batch analysis.
- Experienced in working with various kinds of data sources such as Teradata and Oracle. Successfully loaded files to HDFS from Teradata and loaded from HDFS to Hive and Impala.
- Worked on Oozie workflow engine for job scheduling.
- Experience in setting up the whole app stack, setting up, and debugging Logstash to send Apache logs to AWS Elasticsearch.
- Developed business logic using Kafka Direct Stream in Spark Streaming and implemented business transformations.
- Proven experience with ETL frameworks (Airflow, Luigi, or open-sourced Garcon).
- Developed and maintained batch data flow using HiveQL and Unix scripting.

Environment: Hadoop, AWS, Spark, Micro services, Map Reduce, Agile, HBase, Spark, Kafka, Hive, Pig, Oozie, Tableau.

EDUCATION DETAILS:

Master's: Missouri university of science and technology, Missouri – 2018.

Bachelor's: Teggala Krishna Reddy College of Engineering, Telangana, India – 2015.